

BAYESUVIUS

A VISUAL DICTIONARY OF BAYESIAN
NETWORKS AND CAUSAL INFERENCE



ROBERT R. TUCCI

Bayesuvius,
a visual dictionary of Bayesian Networks and
Causal Inference

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This book is constantly being expanded and improved. To download the latest version, go to <https://github.com/rrtucci/Bayesuvius>

Bayesuvius

by Robert R. Tucci

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Figure 1: View of Mount Vesuvius from Pompeii



Figure 2: Mount Vesuvius and Bay of Naples

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Prologues

Chapter 96

Renormalization Group for Bayesian Networks

In Chapter 95, we gave a general introduction to the Renormalization Group (RG). In Chapter 52, we defined and explored an Ising-dbnet (dbnet= Dynamical Bayesian Network). In this chapter, we propose a bnet RG method applicable to any bnet. As an example, we will apply our bnet RG method to the Ising-dbnet.

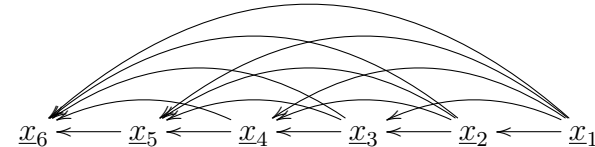
This chapter and its accompanying software `rg4bnet` (see Ref.[103]) are original research, first published here.

Our **bnet RG** method is analogous to the **real space RG** that is used in Condensed Matter Physics. In real space RG, the RG transformation takes a cubic lattice to another cubic lattice with half as many lattice sites. What we propose as an RG transformation for a general bnet $\mathcal{G}$ is to order the nodes $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n$ of $\mathcal{G}$ in topological order, and then to sum over the nodes $\underline{x}_i$ with odd indices i . In conventional real space RG, each RG transformation increases the lattice length by 2. General bnets have no length scale so at first blush, one might think the two methods have little in common. However, note that real space RG and bnet RG both have in common that their RG transformation halves the number of nodes (or lattice sites).

96.1 General bnet RG Method

What we propose as an RG transformation for a general bnet $\mathcal{G}$ is to order the nodes $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n$ of $\mathcal{G}$ in topological order, and then to sum over the nodes $\underline{x}_i$ with odd indices i . We will refer to each halving of nodes as a **decimation step**. The **RG transformation** describes how the coupling constant K changes to K' during a decimation step.

To test our bnet RG method, let us apply it to a bnet $\mathcal{G}$ with six nodes. Suppose bnet $\mathcal{G}$ has six nodes $\underline{x}^{(6)} = (\underline{x}_1, \underline{x}_2, \dots, \underline{x}_6)$, listed in topological order. Then, the chain rule for probabilities states that



$$P(x^6) = \begin{cases} P(x_6 | x_5, x_4, x_3, x_2, x_1) \\ P(x_5 | x_4, x_3, x_2, x_1) \\ P(x_4 | x_3, x_2, x_1) \\ P(x_3 | x_2, x_1) \\ P(x_2 | x_1)P(x_1) \end{cases} \quad (96.1)$$

Of course, Eq.(96.1) is for a fully connected DAG. For a not fully connected DAG, many of the arrows in Eq.(96.1) will be missing.

Next, we perform the following decimation steps

1. Sum over x_5, x_3, x_1 .

This leads to

$$P(x_6, x_4, x_2) = \begin{cases} P(x_6 | x_4, x_2) \\ P(x_4 | x_2) \\ P(x_2) \end{cases} \quad (96.2)$$

Now we have an odd number of nodes. But we need to have an even number of nodes to perform the next decimation step. So we add a dummy node $\underline{x}'_2$ at the beginning of the topological order. If

$$\begin{cases} P(x_2 | x'_2) = \delta(x_2, x'_2) \\ P(\underline{x}'_2 = x'_2) = P(\underline{x}_2 = x'_2) \end{cases} \quad (96.3)$$

then

$$P(x_6, x_4, x_2, x'_2) = \begin{cases} P(x_6 | x_4, x_2) \\ P(x_4 | x_2) \\ P(x_2 | x'_2) \\ P(x'_2) \end{cases} \quad (96.4)$$

2. Sum over x_4 and x'_2 .

This leads to

$$P(x_6, x_2) = \begin{cases} P(x_6 | x_2) \\ P(x_2) \end{cases} \quad (96.5)$$

3. Sum over x_2 and get $P(x_6)$.

This chain of decimation steps is easily generalized to the case where we start with an arbitrary number of nodes. For example, if we start with 50 nodes, the number of nodes at each decimation step would be

$$50 \rightarrow (25 + 1) \rightarrow (13 + 1) \rightarrow (7 + 1) \rightarrow 4 \rightarrow 2 \rightarrow 1 \quad (96.6)$$

In the real space RG of the 2-dim Ising model on a cubic lattice, the RG transformation $K'(K)$ is always approximate, because each decimation generates interactions of each **decimation-surviving (DS)** spin S_{surv} with spins that are farther away from S_{surv} than its 4 nearest-neighbors (NNs). The RG transformation assumes that at the beginning and at the end of the transformation, there are exactly 4 interactions of S_{surv} with its 4 NNs. Hence, one must neglect interactions generated by the RG transformation of S_{surv} with its next to NN spins.

So far, our bnet RG method is exact. However, in practice, we will usually want to make approximations akin to those done in real space RG.

The chain rule for probabilities used above describes a fully connected network wherein the TPM of node i depends on all $i - 1$ previous nodes. However, usually we don't start with a fully connected bnet, and node i has arrows coming into it from fewer than $i - 1$ nodes. In that case, when we decimate, some of the DS nodes may gain incoming arrows.

After each decimation step, the TPM of one or more of the DS nodes might end up with inconveniently many incoming arrows¹. In such cases, we can approximate the TPM of such nodes with a TPM with fewer incoming arrows. Suppose that a DS node $\underline{x}$ has incoming arrows from nodes $\underline{a}^m$ and $\underline{b}^n$, but the interactions with $\underline{b}^n$ are weak and we would like to neglect them. In such a case one can use this approximation:

$$P(x | a^m, b^n) \rightarrow P(x | a^m) = \sum_{b^n} P(x | a^m, b^n) \underbrace{P(b^n | a^m)}_{\approx \prod_{i=1}^n P(b_i)} \quad (96.7)$$

To decide which incoming arrows are weak and can be neglected without introducing too much error, we can use the MI efficiency measure introduced in Chapter 52. Suppose $\underline{x}$ is a DS node and we want to decide whether to neglect or keep arrow $\underline{c} \rightarrow \underline{x}$. Recall from Chapter 52 that the MI efficiency $\epsilon(\underline{x} \leftarrow \underline{c}) \in [0, 1]$ is defined by

$$\epsilon(\underline{x} \leftarrow \underline{c}) = \frac{H(\underline{x} : \underline{c})}{H(\underline{x})} = 1 - \frac{H(\underline{x} | \underline{c})}{H(\underline{x})} \quad (96.8)$$

One strategy one might follow is to specify a threshold $\delta \in [0, 1]$ and neglect arrows $\underline{b}_j \rightarrow \underline{x}$, for all j , that have low efficiency

$$0 \leq \epsilon(\underline{x} \leftarrow \underline{b}_j) < 1 - \delta \quad (96.9)$$

and keep arrows $\underline{a}_i \rightarrow \underline{x}$, for all i , that have high efficiency

$$1 - \delta < \epsilon(\underline{x} \leftarrow \underline{a}_i) \leq 1 \quad (96.10)$$

¹The complexity of the problem increases exponentially with the maximum, over all nodes, of the number of incoming arrows

96.2 Application of bnet RG Method to Ising-dbnet

In this section, we will derive an approximate RG transformation $K'(K)$ for the Ising-dbnet model defined in Chapter 52. As usual, $K = \beta J$, where $\beta = 1/T$ is the inverse temperature and J is the positive spin-spin coupling constant.

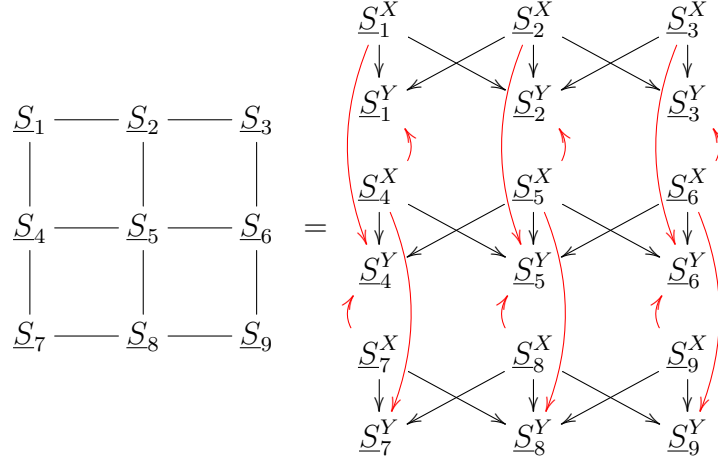


Figure 96.1: DAG definition of 2-dim Ising bnet with 9 dnodes on a 3×3 lattice.

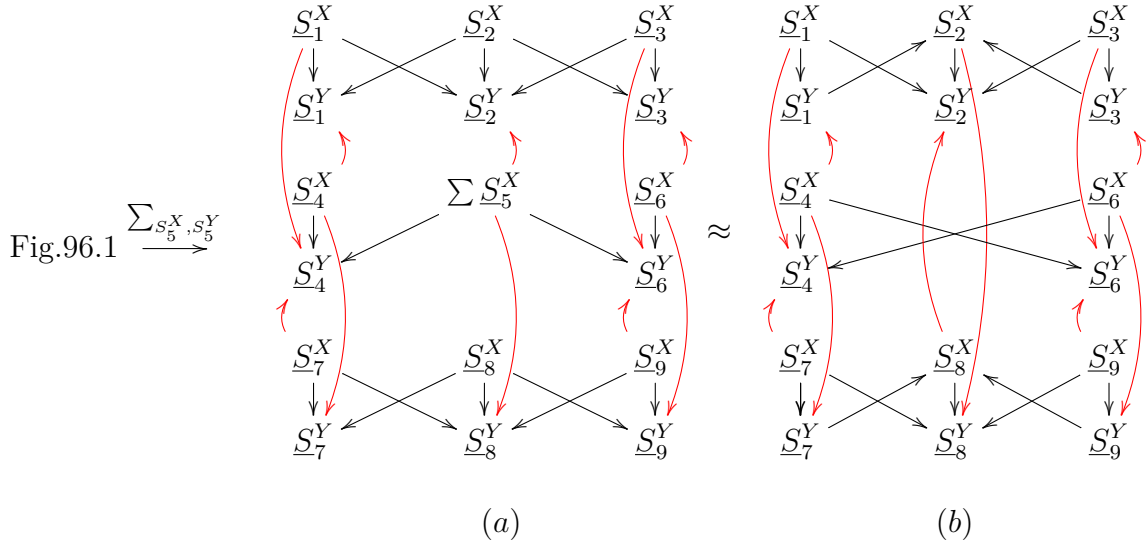


Figure 96.2: After summing over S_5^X and S_5^Y , the bnet in Fig.96.1 becomes the bnet (a). Then that is approximated by the bnet (b).

Consider the bnet of Fig.96.1. Assume that this is the last time slice of an Ising-dbnet. Assume that said Ising-dbnet has gone through enough time slices to “equilibrate”, so that we may safely assume that by the last time slice,

$$S_i^X \approx S_i^Y \quad (96.11)$$

for all i .

Next sum Fig.96.1 over S_5^X and S_5^Y , yielding Fig.96.2 (a). We want to approximate the bnet Fig.96.2 (a) by the bnet Fig.96.2 (b), or, equivalently, we want to approximate the left hand side of Eq.(96.12) by its right hand side.

$$\sum_{S_5^X} P_K(S_5^X) \left\{ \begin{array}{l} P_K(S_4^Y | S_5^X, S_1^X, S_7^X) \\ P_K(S_6^Y | S_5^X, S_3^X, S_9^X) \\ P_K(S_2^Y | S_5^X, S_1^X, S_3^X) \\ P_K(S_8^Y | S_5^X, S_7^X, S_9^X) \end{array} \right\} \approx \left\{ \begin{array}{l} P_{K'}(S_4^Y | S_1^X, S_7^X) \\ P_{K'}(S_6^Y | S_3^X, S_9^X) \\ P_{K'}(S_2^Y | S_1^X, S_3^X) \\ P_{K'}(S_8^Y | S_7^X, S_9^X) \end{array} \right\} \quad (96.12)$$

The conditional probabilities in Eq.(96.12) can be expressed explicitly in terms of exponentials of β times the energy of various spin-spin interactions. For example,²

$$P(S_4^Y | S_5^X, S_1^X, S_7^X) \approx \frac{e^{KS_4^Y(S_5^X + S_1^X + S_7^X)}}{\sum_{S_4^Y} num} \quad (96.13a)$$

$$= \frac{e^{KS_4^Y(S_5^X + S_1^X + S_7^X)}}{2 \cosh[K(S_5^X + S_1^X + S_7^X)]} \quad (96.13b)$$

and

$$P_{K'}(S_4^Y | S_1^X, S_7^X) \approx \frac{e^{K'S_4^Y(S_1^X + S_7^X)}}{\sum_{S_4^Y} num} = \frac{e^{K'S_4^Y(S_1^X + S_7^X)}}{2 \cosh[K'(S_1^X + S_7^X)]} \quad (96.13c)$$

The other conditional probabilities in Eq.(96.12) have explicit expressions analogous to Eqs.(96.13).

Henceforth, we will drop the X or Y superscripts from the spins variables. This is justified because of Eq.(96.11).

Next, let's define the following shorthand notation:

$$e^{Ka} = \left\{ \begin{array}{l} e^{KS_4(S_1 + S_7)} \\ e^{KS_6(S_3 + S_9)} \\ e^{KS_2(S_1 + S_3)} \\ e^{KS_8(S_7 + S_9)} \end{array} \right\} \quad (96.14)$$

² $\sum_i \frac{A_i}{num}$ is shorthand for $\frac{\sum_i A_i}{\sum_i A_i}$. num stands for “numerator”

$$B_K(S_5) = \begin{cases} 2 \cosh[K(S_5 + S_1 + S_7)] \\ 2 \cosh[K(S_5 + S_3 + S_9)] \\ 2 \cosh[K(S_5 + S_1 + S_3)] \\ 2 \cosh[K(S_5 + S_7 + S_9)] \end{cases} \quad (96.15)$$

and

$$\eta = S_2 + S_4 + S_6 + S_8 \quad (96.16)$$

In terms of this shorthand notation, Eq.(96.12) becomes

$$e^{Ka} \sum_{S_5} \frac{P(S_5)}{B_K(S_5)} e^{K\eta S_5} = \frac{e^{K'a}}{B_{K'}(S_5 = 0)} \quad (96.17)$$

If we set

$$a \approx 4 \underbrace{(2 \langle S_1 \rangle)}_{\lambda} \eta = 4\lambda\eta \quad (96.18)$$

and

$$P(S_5 = \pm 1) = P_{\pm} \quad (96.19)$$

then Eq.(96.17) becomes

$$\frac{P_+ e^{K\eta(1+4\lambda)}}{2^4 \cosh^4(K(+1+\lambda))} + \frac{P_- e^{K\eta(-1+4\lambda)}}{2^4 \cosh^4(K(-1+\lambda))} = \frac{e^{K'4\lambda\eta}}{2^4 \cosh^4(K'\lambda)} \quad (96.20)$$

If we divide Eq.(96.20) by itself with $\eta = 0$, we get

$$\frac{1}{\mathcal{N}} \left[\frac{P_+ e^{K\eta(1+4\lambda)}}{\cosh^4(K(+1+\lambda))} + \frac{P_- e^{K\eta(-1+4\lambda)}}{\cosh^4(K(-1+\lambda))} \right] = e^{K'4\lambda\eta} \quad (96.21a)$$

where

$$\mathcal{N} = \frac{P_+}{\cosh^4(K(+1+\lambda))} + \frac{P_-}{\cosh^4(K(-1+\lambda))} \quad (96.21b)$$

From Eqs.(96.21), one finds³ that as $K \rightarrow 0$,

$$K' \rightarrow K + K \frac{P_+ - P_-}{4\lambda} + K^2 \frac{\eta P_+ P_-}{2\lambda} + \mathcal{O}(K^3) \quad (96.22)$$

With $\eta = 4$, $P_- = 3/4$, $\lambda = 1/8$, this becomes

³So as to prevent algebraic errors, all Taylor expansions and asymptotic limits quoted herein were obtained with the symbolic manipulation software SciPy.

$$K' \rightarrow 3K^2 + \mathcal{O}(K^3) \quad (96.23)$$

For those same values of η , P_- and λ , we can also calculate the asymptotic behavior of K' as $K \rightarrow \infty$. We finally obtain, from Eqs.(96.21), that

$$\begin{cases} \text{As } K \rightarrow 0: & K' \rightarrow 3K^2 \\ \text{As } K \rightarrow \infty & K' \rightarrow \frac{5}{2}K - \frac{1}{2}\ln(3) \end{cases} \quad (96.24)$$

One can compare our results for a 2-dim Ising-dbnet to the following ones for the standard 2-dim Ising model. These standard results were obtained via real-space renormalization group (RSRG) using a block-spin decimation approximation (BSDA).

$$\begin{cases} K'(K) = \frac{3}{8} \ln [\cosh(4K)] \\ \text{As } K \rightarrow 0: & K' \rightarrow 3K^2 \\ \text{As } K \rightarrow \infty: & K' \rightarrow \frac{3}{2}K - \frac{3}{8} \ln 2 \end{cases} \quad (96.25)$$

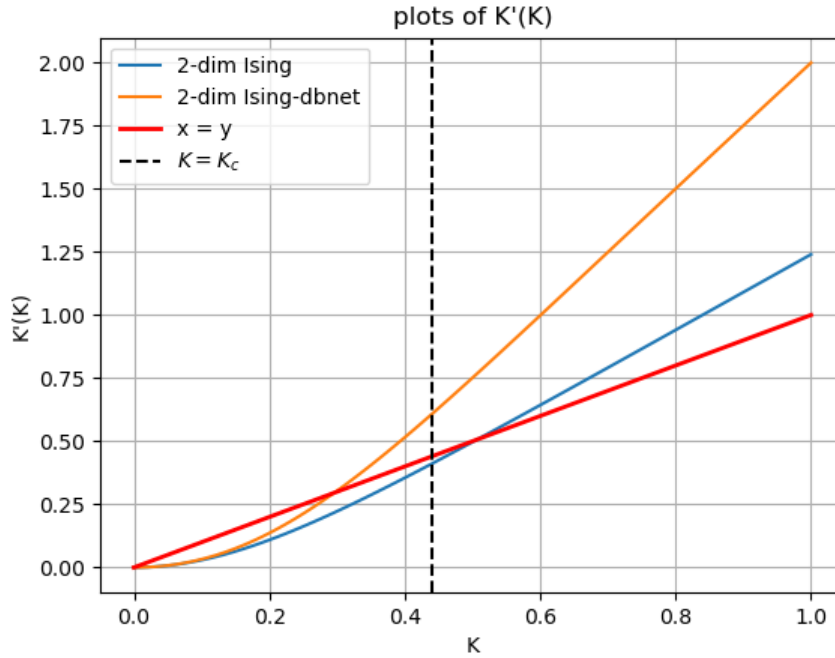


Figure 96.3: Plot of RG transformations $K'(K)$ for the standard 2-dim Ising model in the block-spin decimation approximation, and for the 2-dim Ising-dbnet model. According to the exact Onsager solution, the Curie critical point should be at $K_c = \beta_c J = J/T_c \approx 0.44$.

See Fig.96.3 for a plot of the RG transformations: Eqs.(96.21) for the Ising-dbnet and Eq.(96.25) for the standard Ising model. In both cases, one observes a

repulsive fixed point located close to, but not exactly at, the position predicted by the exact Onsager solution.

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